

THE ROLE OF ASSISTIVE TECHNOLOGY IN SUPPORTING THE LEARNING AND PARTICIPATION OF STUDENTS WITH PHYSICAL CHALLENGES IN SELECTED SECONDARY SCHOOLS IN OYO STATE, NIGERIA

Emmanuel, Mary

Department of Guidance and Counselling Education, Ekiti State University, Ado-Ekiti

In Affiliation with Emmanuel Alayande University of Education, Oyo, Nigeria

Supervisor: Assoc. Prof. Dr. A.O. Oyediran | Submitted: May 2026

ABSTRACT

This study investigated the role of assistive technology in supporting the learning and participation of students with physical challenges in selected secondary schools in Oyo State, Nigeria. Anchored on the Technology Acceptance Model (TAM) and the Universal Design for Learning (UDL) framework, the study employed a descriptive survey design. The sample comprised 150 respondents—90 teachers and 60 physically challenged students—selected through stratified random sampling. Data were collected via a structured, researcher-developed questionnaire (Cronbach's $\alpha = 0.86$) and analyzed using descriptive statistics and Chi-square tests of independence ($\alpha = 0.05$). Findings revealed a severe shortage of assistive technologies, with over 70% of respondents reporting unavailability across all device categories. Despite this, 62–76% of respondents agreed that assistive technology enhances academic performance, and 74–79% affirmed its positive impact on classroom participation and social inclusion. Chi-square analysis found no statistically significant relationships between assistive technology availability and academic performance ($X^2 = 0.173$, $p = 0.917$), classroom participation ($X^2 = 3.014$, $p = 0.222$), or social inclusion ($X^2 = 0.978$, $p = 0.613$)—an outcome attributable to the near-universal absence of devices rather than their ineffectiveness. Private and urban schools had significantly better access than public and rural counterparts. Key barriers included inadequate funding, insufficient teacher training, and poor infrastructure. The study recommends increased government investment in assistive technology, integration of inclusive technology training in teacher education, and stronger policy enforcement to advance equitable education.

Keywords: Assistive Technology, Physical Challenges, Inclusive Education, Academic Performance, Classroom Participation, Nigeria

1. INTRODUCTION

Education is universally acknowledged as a fundamental human right and a cornerstone of individual empowerment, social mobility, and national development (UNESCO, 2023). For learners with physical challenges, however, achieving equitable access to education remains a formidable task, particularly in developing countries such as Nigeria, where structural, social, and technological barriers persist across schools (Ajuwon & Chukwuma, 2020).

Assistive technology (AT) has emerged globally as a powerful enabler for learners with disabilities, encompassing a wide spectrum—from low-tech solutions such as walking aids and magnifiers to high-tech innovations including speech-to-text software, electronic Braille displays, and motorized wheelchairs (Sharma & Singh, 2021). In education, these tools promote independence, active classroom participation, and meaningful social inclusion (Olowe & Dada, 2020). Yet in Nigeria, the adoption of assistive technology in schools remains highly inadequate due to prohibitive costs, limited educator awareness, infrastructural deficits, and poor technical maintenance (Eze et al., 2021; Oni, 2019).

This study is motivated by a critical gap in localized empirical evidence on how assistive technologies are deployed and experienced in Nigerian inclusive secondary schools, and what barriers continue to constrain their adoption. By investigating the types of technologies available, their perceived impact on academic performance and classroom participation, and the challenges limiting their effectiveness, this research aims to provide evidence-based insights for educators, policymakers, and stakeholders toward advancing inclusive, equitable education in Nigeria.

1.1 Objectives of the Study

The general objective was to investigate the role of assistive technology in supporting the learning and participation of students with physical challenges in secondary schools in Oyo State, Nigeria. Specific objectives were to:

- Identify the types of assistive technologies available to physically challenged students in selected schools.
- Examine the extent to which assistive technology contributes to academic performance.
- Evaluate the impact of assistive technology on classroom participation and social inclusion.
- Explore the challenges hindering the effective use of assistive technology.
- Propose strategies for improving the adoption and integration of assistive technology in inclusive education.

1.2 Research Hypotheses

- H_{01} : There is no significant relationship between the use of assistive technology and the academic performance of students with physical challenges.
- H_{02} : There is no significant relationship between the use of assistive technology and the classroom participation of students with physical challenges.
- H_{03} : There is no significant relationship between the use of assistive technology and the social inclusion of students with physical challenges.

2. LITERATURE REVIEW

2.1 Concept and Scope of Assistive Technology

Assistive technology encompasses devices, systems, and applications that enhance the functional abilities of individuals with disabilities, enabling them to perform tasks that would otherwise be difficult or impossible (WHO, 2022). In educational contexts, AT facilitates learner engagement, curriculum access, and meaningful peer interaction. Researchers classify these tools into three broad categories: low-tech (e.g., tactile maps, Braille books, modified seating), mid-tech (e.g., audio recorders, adapted keyboards), and high-tech (e.g., screen readers, speech-to-text software, digital Braille displays) (Edeh et al., 2022).

In Nigeria, while AT is increasingly recognized as vital for inclusive education, its adoption is constrained by financial, infrastructural, and attitudinal barriers (Ajuwon & Chukwuma, 2020). Public schools are largely ill-equipped, and students with physical challenges often resort to peer support and teacher assistance rather than technological solutions, reinforcing dependency and educational inequality (Okeke & Agbo, 2021).

2.2 Inclusive Education and Physical Challenges

Inclusive education represents a transformative approach ensuring every learner, regardless of physical or other differences, has equal access to quality education within the same learning environment (UNESCO, 2020).

Nigeria's National Policy on Inclusive Education (FME, 2018) defines inclusion broadly, yet implementation remains inconsistent; many schools still operate under exclusionary structures where learners with disabilities lack necessary support (Okeke & Agbo, 2021).

Physical challenges encompass a wide range of conditions—including cerebral palsy, spinal cord injuries, muscular dystrophy, spina bifida, and limb deficiencies—that can restrict access, participation, and independent engagement with learning materials (WHO, 2022). These conditions do not inherently impair cognitive ability; rather, it is the failure of educational environments to accommodate diversity that constitutes the real barrier (Florian & Spratt, 2019; Oliver, 1996).

2.3 Theoretical Framework

This study is anchored on two complementary frameworks:

The Technology Acceptance Model (TAM), developed by Davis (1989), posits that two key factors determine technology adoption: perceived usefulness and perceived ease of use. In Nigerian school contexts, low digital literacy among teachers, unstable electricity, poor internet connectivity, and limited institutional support collapse perceived ease of use, suppressing AT adoption even when devices are available (Eze et al., 2021; Nwosu & Eke, 2020).

The Universal Design for Learning (UDL) framework (Meyer, Rose, & Gordon, 2014) advocates for designing flexible educational environments that accommodate individual learning differences from the outset—through multiple means of representation, engagement, and expression. UDL and AT are complementary: UDL provides the inclusive structural framework, while AT personalizes the learning experience for each student (UNESCO, 2023).

2.4 Empirical Review

Globally, empirical studies consistently affirm AT's positive impact. Alnahdi (2020) found that AT improved academic participation by 47% among students with disabilities. Ok, Kim, and Rao's (2021) meta-analysis of 48 studies reported significant improvements in academic performance and social interaction when AT was integrated in inclusive classrooms. In Africa, Mtebe and Raisamo (2020) found that South African students who used adaptive technologies performed significantly better in written communication, while Mpofu and Chitiyo (2022) reported that Zimbabwean students with AT access showed higher academic confidence and engagement.

In Nigeria, Ajuwon and Chukwuma (2020) found that while 73% of teachers recognized AT's value, only 28% used it regularly. Olowe and Dada (2020) demonstrated a 35% increase in physically challenged students' classroom inclusion following a two-week teacher training program. Akinola and Oke (2023) reported significant improvements in cognitive engagement and literacy outcomes among Oyo State students using adaptive e-learning software. These findings collectively highlight the transformative potential of AT, constrained primarily by systemic rather than individual factors.

3. METHODOLOGY

3.1 Research Design and Study Area

The study adopted a descriptive survey research design, which enables the collection of data from a representative sample to describe existing conditions and attitudes without manipulating variables (Creswell & Creswell, 2018). The study was conducted in selected secondary schools across Oyo State, Nigeria—a geographically diverse setting encompassing both urban and rural institutions and a mix of public and private schools, allowing for comparative analysis.

3.2 Sample and Sampling Technique

The sample comprised 150 respondents: 90 teachers and 60 physically challenged students. Stratified random sampling was applied—schools were first stratified by type (public/private) and location (urban/rural), and participants were then randomly selected within each stratum to ensure representative coverage of Oyo State's heterogeneous educational landscape (Kothari, 2018).

3.3 Research Instrument

Data were collected using a structured, researcher-developed questionnaire divided into six sections: demographic information; availability of assistive technology; effect of AT on academic performance; impact on classroom participation and social inclusion; challenges hindering AT use; and strategies for improving AT integration. Items in Sections B–F used a four-point Likert scale (Strongly Agree to Strongly Disagree). The instrument was validated by three specialists from the University of Ibadan (educational psychology, measurement and evaluation, and special education), and its reliability was established through pilot testing ($n = 20$), yielding a Cronbach's Alpha of 0.86—well above the 0.70 threshold (Nunnally & Bernstein, 1994).

3.4 Method of Data Analysis

Data were analyzed using SPSS. Descriptive statistics (frequencies, percentages) were applied to summarize demographic distributions and general response patterns. Chi-square tests of independence were used to test the three research hypotheses at the 0.05 significance level. Respondents' Likert scores were summed across relevant items to produce composite availability, academic performance, classroom participation, and social inclusion scores, which were then categorized for cross-tabulation.

4. RESULTS AND ANALYSIS

4.1 Demographic Profile of Respondents

The 150 respondents were evenly distributed by gender (50% male, 50% female), comprising 90 teachers (60%) and 60 physically challenged students (40%). The sample was balanced across school type (52% public, 48% private) and location (51.3% rural, 48.7% urban), with diverse experience levels represented.

4.2 Availability of Assistive Technologies

Descriptive analysis revealed a severe deficiency in AT availability across the sampled schools. Table 1 summarizes agreement levels for availability-related statements.

Statement	Agree/Strongly Agree	Disagree/Strongly Disagree
Physical devices (wheelchairs, ramps, adjustable desks)	20.0%	80.0%
Computer-based tools (screen readers, voice-to-text)	26.7%	73.3%
Accessible learning materials (Braille, audio lessons)	20.0%	80.0%
Infrastructural support (elevators, accessible restrooms)	17.3%	82.7%
Teacher/student awareness of available AT	17.3%	82.7%

Table 1: Availability of Assistive Technologies ($n = 150$)

4.3 Effect of Assistive Technology on Academic Performance

Despite the alarming shortage, respondents expressed overwhelmingly positive perceptions of AT's educational value. Between 62.7% and 76.0% agreed that AT helps students understand lessons better, complete academic tasks independently, perform better in examinations, and improve concentration. This perception paradox—widespread belief in AT's effectiveness coexisting with near-universal unavailability—is the central finding of this study.

4.4 Impact on Classroom Participation and Social Inclusion

Agreement rates for AT's positive impact on participation and inclusion ranged from 74.7% to 79.3% across five statements, including: enabling full classroom discussion participation (74.7%), encouraging inclusion in group work and extracurricular activities (79.3%), and reducing discrimination and social isolation (78.7%). Respondents thus recognize AT as both an academic tool and a social equalizer.

4.5 Challenges Hindering Effective AT Use

Five major systemic barriers were identified, each recording over 70% disagreement with positive statements:

Challenge	% Reporting as a Barrier
Inadequate government funding	70.7%
Insufficient teacher training	72.0%
Lack of technical support	71.3%
Unstable electricity and internet access	71.3%
Weak management commitment to inclusive education	73.3%

Table 2: Challenges Hindering Effective AT Use (n = 150)

4.6 Strategies for Improving AT Integration

Respondents strongly endorsed five strategies, with agreement levels ranging from 86.0% to 90.0%:

Strategy	% Agreement
Increased government funding for inclusive education	90.0%
Policy mandate for AT in all inclusive schools	90.0%
Partnerships with NGOs and technology companies	89.3%
Regular teacher training programs on AT	86.0%
Awareness campaigns for inclusive attitudes	86.0%

Table 3: Strategies for Improving AT Integration (n = 150)

4.7 Hypothesis Testing (Chi-Square Analysis)

Chi-square tests of independence were conducted to test the three null hypotheses at $\alpha = 0.05$.

Hypothesis	X ²	df	p-value	Decision
H ₀₁ : AT Availability × Academic Performance	0.173	2	0.917	Not Rejected

Hypothesis	X ²	df	p-value	Decision
H ₀₂ : AT Availability × Classroom Participation	3.014	2	0.222	Not Rejected
H ₀₃ : AT Availability × Social Inclusion	0.978	2	0.613	Not Rejected
School Type × AT Availability	4.089	1	0.043*	Significant
School Location × AT Availability	7.361	1	0.007*	Significant
Teacher/Student × Performance Perceptions	0.899	2	0.638	Not Significant

Table 4: Chi-Square Results Summary (* $p < 0.05$)

None of the three primary null hypotheses were rejected. The non-significant results for H₀₁, H₀₂, and H₀₃ do not indicate that AT lacks educational value; rather, they reflect the statistical reality that near-universal absence of AT across sampled schools prevents measurable differences in outcomes from emerging. Significant associations were, however, established between school type and AT availability ($p = 0.043$) and school location and AT availability ($p = 0.007$), confirming that private and urban schools possess significantly better AT resources than public and rural counterparts.

5. DISCUSSION

5.1 The AT Availability Crisis

The documentation of severe AT shortages in Oyo State secondary schools—particularly across public and rural institutions—mirrors national trends reported by Oni (2019) and Okeke and Agbo (2021), and reflects the global inequity identified by the WHO (2022), wherein only one in ten individuals requiring assistive products has access to them. Nigeria's ratification of the CRPD and enactment of the National Policy on Inclusive Education (FME, 2018) have not translated into meaningful resource provision at the school level. The TAM framework (Davis, 1989) helps explain this gap: without foundational enabling conditions such as stable electricity, technical support, and teacher training, even the rare instances where AT exists cannot drive sustainable adoption.

5.2 Perceptions versus Statistical Reality

The paradox between high perceptual agreement and statistically non-significant hypothesis outcomes is the most theoretically significant contribution of this study. The findings confirm TAM's assertion that perceived usefulness alone is insufficient to drive outcomes; actual usage, enabled by consistent availability and institutional support, is the necessary condition (Venkatesh & Davis, 2000). The elevated chi-square value for H₀₂ (3.014) relative to H₀₁ (0.173) suggests that participation effects begin to emerge even under conditions of limited availability, consistent with UTAUT theory (Venkatesh et al., 2003).

5.3 Barriers and Solutions: A Systemic Diagnosis

The five major barriers identified—inadequate funding, insufficient teacher training, lack of technical support, infrastructure instability, and weak management commitment—are interconnected and mutually reinforcing. The challenge of teacher training is particularly acute: Olowe and Dada (2020) demonstrated that a two-week training program produced a 35% increase in classroom inclusion of physically challenged students, underscoring what is currently being foregone. The UDL framework (Meyer, Rose, & Gordon, 2014) further

emphasizes that without educators trained to embed AT proactively into lesson design, devices remain equipment rather than pedagogy.

The near-universal endorsement of government funding (90.0%) and policy mandates (90.0%) as strategies reflects stakeholders' understanding that systemic change requires state-level commitment grounded in the human rights-based model of disability (United Nations, 2006). The Discrimination Against Persons with Disabilities (Prohibition) Act (2018) provides a legal foundation, but enforcement mechanisms remain weak.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This study demonstrates that assistive technology holds transformative—and well-recognized—potential for enhancing the learning, participation, and social inclusion of students with physical challenges in Nigerian secondary schools. The critical barrier is not awareness or willingness but the severe, systemic unavailability of AT tools, rooted in inadequate funding, limited teacher training, poor infrastructure, and weak institutional commitment. The non-rejection of all three hypotheses encapsulates Nigeria's inclusive education paradox: a resource widely acknowledged as effective yet overwhelmingly absent for those who need it most.

Nigeria cannot achieve SDG 4 or honor its CRPD obligations without making AT integration into its inclusive education system a national priority. Every physically challenged student denied access to these tools is denied not merely an assistive device, but an education, a future, and a right.

6.2 Recommendations

Based on the findings, the following recommendations are advanced:

- The Federal and State Governments should significantly increase budgetary allocations for AT procurement in secondary schools, prioritizing public and rural institutions, with independent monitoring of fund utilization.
- The Federal Ministry of Education should mandate a comprehensive teacher training program on AT and inclusive pedagogy as a compulsory component of both pre-service and continuous professional development.
- Schools should formalize partnerships with NGOs, technology companies, and international development agencies, with clear agreements on device provision, maintenance, and technical support to ensure sustainable outcomes.
- Regular awareness campaigns should be conducted targeting teachers, administrators, parents, and communities to promote rights-based understanding of disability and challenge cultural stigmas.
- The National Policy on Inclusive Education should be revised to explicitly mandate minimum AT standards in all inclusive schools, with constructive compliance monitoring through school inspection frameworks.
- Guidance counselors should be formally empowered to identify AT needs, advocate for appropriate devices, coordinate teacher integration strategies, and provide psychosocial support to students with disabilities.
- Infrastructure planning for schools should incorporate Universal Design for Learning principles, including ramps, accessible restrooms, adjustable furniture, reliable electricity, and internet connectivity.
- Local researchers and technology institutions should be supported through grants and policy incentives to develop affordable, context-specific assistive devices suited to Nigerian realities.

6.3 Suggestions for Further Studies

- Experimental or quasi-experimental designs should be employed in future studies to establish causal relationships between specific AT interventions and learning outcomes.
- A nationwide multi-state replication is recommended to establish whether Oyo State findings are generalizable across Nigeria.
- Mixed-method approaches combining surveys with interviews, focus groups, and classroom observations would provide richer, more contextual understanding of students' lived experiences with AT.
- Longitudinal studies tracking academic trajectories of physically challenged students with and without consistent AT access would provide robust evidence for sustained public investment.

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